

Bacterial Cell Cycle Regulation: A Hierarchical Framework for Physical-Molecular Integration

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Abstract

The bacterial cell cycle involves chromosome replication, segregation, and division. We propose a hierarchical framework for understanding how physical constraints and molecular regulation interact, distinguishing three organizational types: Type A (Hierarchical Override) where molecular regulation dominates during checkpoints and stress responses; Type B (Bidirectional Coupling) where physical and molecular systems interact continuously during homeostasis; and Type C (Physical Default) where physical processes dominate when molecular regulation is minimal. We introduce the Asymmetry Index ($AsI = |\Delta M / \sigma M| / |\Delta P / \sigma P|$) for quantifying molecular versus physical influences.

This framework provides an integrated perspective: physical constraints and molecular regulation are complementary rather than competing, with relationships depending on functional context. Molecular systems **operate within** physical constraints during homeostasis (Type B), **override** physical constraints during critical transitions (Type A), and **rely on** physical defaults when regulatory complexity is minimal (Type C).

1. Introduction

1.1 The Bacterial Cell Cycle: A Multi-Level Regulatory Problem

The bacterial cell cycle consists of three coordinated processes: chromosome replication, segregation, and division. Classical molecular biology has identified numerous regulatory proteins forming sophisticated control circuits. In *Escherichia coli*, replication initiation involves DnaA, DnaC, DiaA, SeqA, Dam methylase, RIDA, datA locus, and DARS sequences (Mott & Berger, 2007; Katayama et al., 2017; Nishida et al., 2022). Chromosome segregation employs ParA/ParB systems and SMC complexes (Di Lallo et al., 2003; Wang et al., 2017; Le Gall et al., 2022). Division requires FtsZ, FtsA, ZipA, ZapA, MinCDE system, and nucleoid occlusion factors (Adler et al., 1967; de Boer et al., 1989; Bernhardt & de Boer, 2005; Rivas et al., 2022).

The prevailing view frames these as evolved regulatory circuits ensuring coordination through molecular feedback loops (Moolman et al., 2014; Shi et al., 2018), with physical constraints acknowledged as boundary conditions but not primary determinants.

1.2 The Fundamental Question

To what extent do physical and chemical constraints contribute to bacterial cell cycle regulation, and how do these physical foundations interact with sophisticated molecular regulatory systems? This question has implications for understanding cellular organization, reconstructing early cellular evolution, designing minimal synthetic cells, and distinguishing between physical constraints and molecular regulation.

1.3 The Hierarchical Framework: Key Insights

Rather than asking "physical versus molecular regulation" as a binary choice, we ask "**When do molecular systems override physical constraints, and when do they work within them?**" Asymmetric information flow—molecular regulation overriding physical constraints—emerges during critical functional transitions: stress responses (SOS response), developmental programming (*Caulobacter* asymmetric division), and checkpoint activation (Janion, 2008; Baharoglu & Mazel, 2014). During normal homeostasis, many systems exhibit bidirectional coupling where physical and molecular systems influence each other continuously (Murray, 2004; Liu & Wang, 1987).

This perspective transforms our understanding of cellular life. For origins-of-life research, physical constraints may provide a framework for early cellular organization. For synthetic biology, understanding which functions require molecular sophistication guides minimal cell design (Breuer et al., 2019). For evolutionary biology, distinguishing between physically constrained traits and contingent evolutionary solutions reveals what aspects of cellular organization are inevitable versus chosen.

1.4 Scope and Limitations

This review focuses primarily on well-studied model systems (*E. coli*, *B. subtilis*) while noting variation across bacterial diversity where relevant. We acknowledge that our understanding continues to evolve and that some areas remain speculative with limited direct evidence.

2. Physical Constraints: The Foundational Context

This section reviews physical and chemical constraints that create permissive conditions for cell cycle progression. **We emphasize throughout that molecular regulation is essential for achieving the precision and robustness observed in real bacteria**—physical constraints set boundary conditions, but molecular systems determine actual outcomes.

Important caveat on evidential strength: The physical constraints discussed vary substantially in evidential support. Nucleoid geometry and DNA topology have well-established molecular mechanisms linking physical parameters to regulatory outcomes. Turgor pressure and macromolecular crowding show correlations but causal mechanisms remain uncertain.

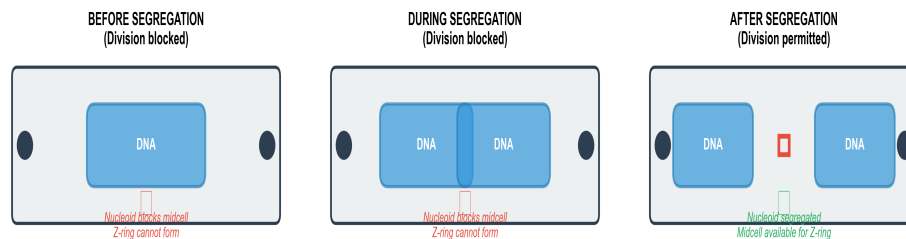
2.1 Nucleoid Geometry: Spatial Constraints on Division Placement

Nucleoid geometry constrains where division can occur through two well-established mechanisms: nucleoid occlusion and the Min system. Nucleoid occlusion prevents Z-ring formation over nucleoid material, ensuring division occurs only after chromosome segregation (Bernhardt & de Boer, 2005; Wu

& Errington, 2004; Wu et al., 2016; Rivas & Margolin, 2018). In *E. coli*, SlmA binds specific DNA sequences and prevents FtsZ polymerization over unsegregated nucleoids (Bernhardt & de Boer, 2005; Tonthat et al., 2017). In *B. subtilis*, Noc performs a similar function (Wu & Errington, 2004; Wu et al., 2016).

The Min system prevents polar divisions by oscillating between cell poles and inhibiting Z-ring formation everywhere except midcell (de Boer et al., 1989; Raskin & de Boer, 1997; Hu & Lutkenhaus, 1999; Meacci & Kruse, 2005; Huang et al., 2024). The Min system's self-organization is well-characterized both experimentally and theoretically, with recent work showing that Min patterns adapt to cell shape changes (Huang et al., 2024).

Nucleoid Occlusion: Physical Constraint Translated to Molecular Regulation



MOLECULAR SENSING & RESPONSE: Physical Constraint (nucleoid at midcell) → Molecular Detection (SlmA/Noc/Min) → Functional Outcome (division at proper time/place)

Figure 1. Nucleoid occlusion and Min system spatial constraints. The nucleoid (blue) occupies the cellular volume, preventing Z-ring formation (red) over unsegregated chromosomes. The Min system (green oscillations) prevents polar divisions, ensuring midcell Z-ring placement.

These mechanisms demonstrate that bacterial cells actively sense and respond to geometric and topological constraints. **However**, the exact relationship between nucleoid geometry, Min system behavior, and division placement remains an area of active research.

2.2 DNA Topology: Supercoiling as a Regulatory Signal

DNA supercoiling—the twisting and coiling of the DNA double helix—affects replication, transcription, and chromosome segregation (Liu & Wang, 1987; Postow et al., 2001; Dorman, 2013; López-García et al., 2021). Negative supercoiling promotes DNA strand separation, facilitating replication initiation and transcription. Topoisomerases regulate supercoiling levels, creating a dynamic balance between underwound and overwound DNA states.

Supercoiling levels correlate with growth phase and metabolic state, suggesting a regulatory role (Dorman, 2013; Blumenthal et al., 2020). During rapid growth, increased negative supercoiling facilitates replication initiation. During stress or starvation, reduced supercoiling may slow replication and conserve resources.

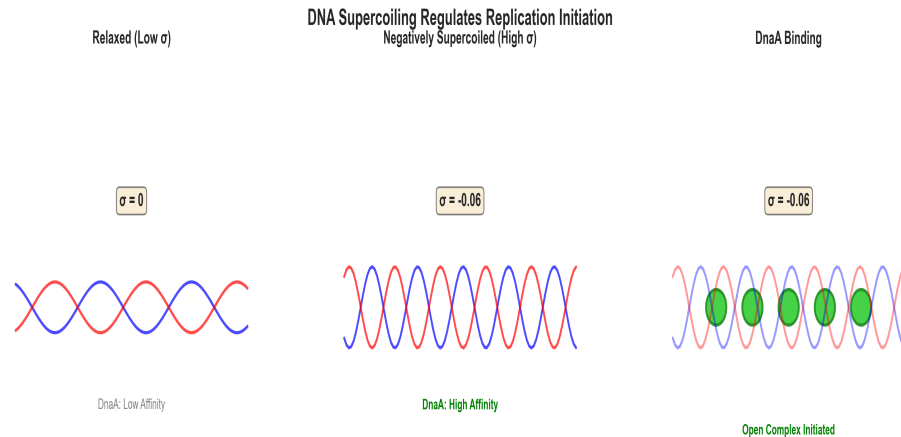


Figure 2. DNA supercoiling as a bidirectional regulatory signal. Negative supercoiling promotes DNA strand separation for replication and transcription. The relationship between supercoiling and cell cycle progression is bidirectional: DNA topology affects replication and transcription, while replication and transcription alter DNA topology.

The relationship between supercoiling and cell cycle progression appears to be **bidirectional**: DNA topology affects replication and transcription, while replication and transcription alter DNA topology (Liu & Wang, 1987; Dorman, 2013). This bidirectional coupling exemplifies how physical and molecular systems can interact continuously during normal homeostasis.

2.3 Turgor Pressure and Cell Size Homeostasis

Turgor pressure—the outward force exerted by the cytoplasmic membrane against the cell wall—increases as bacterial cells grow due to the surface-to-volume ratio (Huang et al., 2013; Zhou et al., 2023). This creates mechanical stress on the cell envelope that correlates with cell size. Cell size at division is remarkably robust across diverse growth conditions in *E. coli* and *B. subtilis* (Shi et al., 2018; Witz et al., 2019).

The "adder" principle has emerged as the dominant model for bacterial size homeostasis: cells add a constant size increment between divisions regardless of initial size (Campos et al., 2014; Taheri-Araghi et al., 2015). Amir (2014) showed that adder behavior can emerge from stochastic molecular processes combined with threshold triggering. Multiple molecular mechanisms have been proposed, including FtsZ accumulation-based models (Moolman et al., 2014; Si et al., 2019) and replication-division coupling models (Si & Levin, 2020; Wallden et al., 2016).

The adder principle likely represents Type B bidirectional coupling: molecular systems (FtsZ accumulation) and physical parameters (cell growth) are continuously coupled to maintain homeostasis. The fact that adder behavior varies across conditions and species underscores the context-dependent nature of physical-molecular relationships.

2.4 Macromolecular Crowding and Entropic Forces

The bacterial cytoplasm is densely packed with macromolecules, with estimated concentrations of 300-400 mg/mL (Minton, 2000; Zhou et al., 2008; Ellis, 2001). This crowding creates excluded volume effects that favor compact molecular conformations and enhance association reactions (Minton, 2000). Crowding has been implicated in protein folding, complex formation, and phase separation of biomolecular condensates (Shin & Brangwynne, 2017; Guillen-Boixet et al., 2020).

However, **direct evidence linking crowding to cell cycle regulation remains limited**. Most studies show that crowding affects molecular behavior in vitro, but whether crowding acts as a specific regulatory signal for cell cycle transitions in vivo is unclear.

3. Molecular Regulation: Essential and Sophisticated

Physical constraints create boundary conditions, but molecular regulation is essential for achieving the precision, robustness, and adaptability observed in bacterial cells.

3.1 Replication Initiation: Multiple Overlapping Control Layers

Replication initiation in *E. coli* is regulated by multiple overlapping mechanisms ensuring precise timing and coordination. DnaA, the initiator protein, binds to oriC and unwinds DNA to initiate replication (Mott & Berger, 2007; Katayama et al., 2017). DnaA activity is regulated by ATP/ADP binding, with DnaA-ATP being active for initiation (Sekimizu et al., 1987; Nishida et al., 2022). The DnaA-ATP/DnaA-ADP ratio is controlled by RIDA (Regulatory Inactivation of DnaA), DARS (DnaA Reactivating Sequences), and datA locus (Katayama et al., 2017; Kasho & Katayama, 2022; Kono & Katayama, 2021).

Additional regulators include SeqA, which sequesters hemi-methylated oriC after replication to prevent re-initiation (Campbell & Kleckner, 1990; Landoulsi et al., 2021), and DiaA, which stimulates DnaA assembly (Ishida et al., 2004; Keyamura et al., 2007).

RIDA as a Type B system: The Regulatory Inactivation of DnaA (RIDA) system exemplifies Type B bidirectional coupling. RIDA couples a physical parameter—replication fork progression—to molecular regulation of DnaA activity. As the replication fork advances, the sliding clamp (β -clamp) loads Hda protein, which stimulates DnaA-ATP hydrolysis, converting active DnaA-ATP to inactive DnaA-ADP (Katayama et al., 1998; Kono & Katayama, 2021). This creates continuous feedback: the physical process of DNA replication directly regulates the molecular initiator.

Metabolic coordination: The alarmone ppGpp (guanosine tetraphosphate) provides another critical physical-molecular interface, linking metabolic state to cell cycle progression. During nutrient limitation, ppGpp accumulates and directly inhibits replication initiation by binding to DnaA and reducing its affinity for oriC (Battesti & Bouveret, 2006; Gourse et al., 2018). This system demonstrates Type B bidirectional coupling at the organismal level.

3.2 Chromosome Segregation: Active and Passive Mechanisms

Chromosome segregation involves both active and passive mechanisms. ParA/ParB systems actively pull chromosomes apart using ATP-dependent mechanisms (Di Lallo et al., 2003; Ringgaard et al.,

2009; Le Gall et al., 2022). SMC complexes organize and condense chromosomes (Wang et al., 2017; Bürmann et al., 2021; Nolivos et al., 2022). DNA replication and transcription also contribute to segregation through passive mechanisms (Dworkin & Losick, 2002; Bates & Maxwell, 2005).

3.3 Division Septum Formation: Spatial and Temporal Control

Division septum formation requires precise spatial and temporal control. In most bacteria and many archaea, FtsZ polymerizes into a Z-ring at midcell, providing the scaffold for divisome assembly (Adler et al., 1967; Bi & Lutkenhaus, 1991; Huang et al., 2024). FtsA and ZipA anchor FtsZ to the membrane (Pichoff & Lutkenhaus, 2002). The Min system ensures proper Z-ring placement by inhibiting FtsZ polymerization at cell poles (de Boer et al., 1989; Raskin & de Boer, 1997). Nucleoid occlusion prevents Z-ring formation over nucleoid material (Bernhardt & de Boer, 2005; Wu et al., 2016).

3.3.1 Alternative Division Systems: Evolutionary Plasticity

While FtsZ-based division is the most widespread mechanism, diverse prokaryotes use alternative division systems. Many archaea lack FtsZ entirely and use ESCRT-III machinery or Crenactin-based systems for division (Samson et al., 2022; Lindås et al., 2008; Ettema et al., 2011). Some bacterial groups have also lost FtsZ: *Chlamydia trachomatis* divides using unknown mechanisms, and Planctomycetes reproduce by budding rather than binary fission (Stephens et al., 1998; Jogler et al., 2012).

The diversity of division systems reveals that physical constraints (membrane geometry, turgor pressure) are permissive rather than deterministic—multiple molecular mechanisms can achieve the same physical outcome. Organisms lacking FtsZ may operate closer to Type C physical-default organization, though this requires experimental validation using AsI measurements.

4. The Hierarchical Framework and Asymmetry Index

4.1 Three Types of Physical-Molecular Relationships

Type A: Hierarchical Override—Molecular regulation dominates physical processes during critical functional transitions: checkpoints, stress responses, and developmental programming.

Canonical example: The SOS DNA damage checkpoint blocks division despite permissive physical conditions (Janion, 2008; Baharoglu & Mazel, 2014). When DNA is damaged, the SOS response upregulates DNA repair genes and inhibits cell division via SulA, regardless of cell size, turgor pressure, or other physical conditions that would normally permit division.

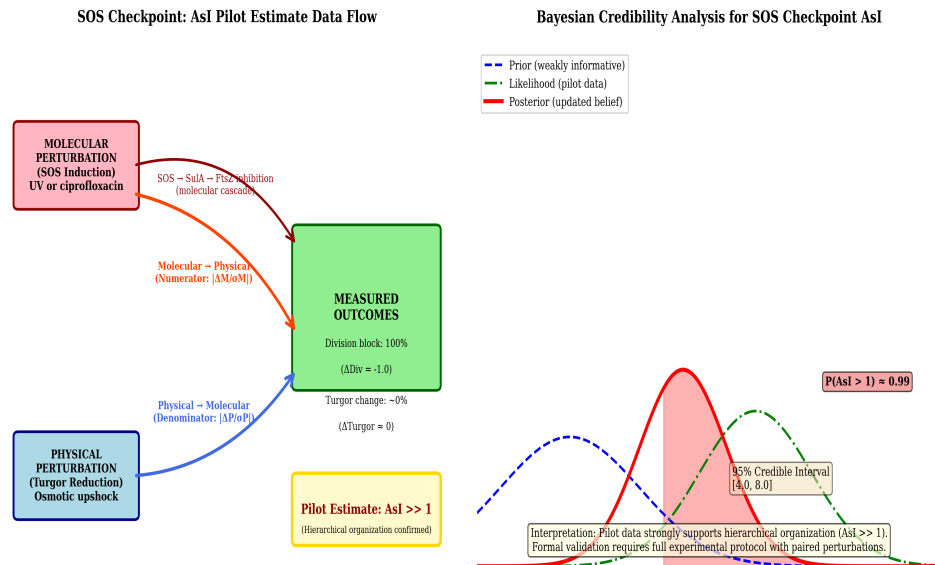


Figure 3. Type A Hierarchical Override: The SOS DNA damage checkpoint. When DNA damage is detected (left), the SOS response activates molecular inhibitors (SulA) that block cell division (right), regardless of permissive physical conditions. This demonstrates molecular regulation overriding physical constraints during stress responses.

**Canonical example 2: Caulobacter crescentus developmental programming*—Caulobacter* exhibits an obligate asymmetric cell cycle producing morphologically and functionally distinct daughter cells: a motile swarmer cell and a sessile stalked cell (Laub et al., 2000; Shapiro et al., 2002). This developmental asymmetry is orchestrated by a hierarchical molecular regulatory network centered on the CtrA response regulator.

CtrA~P (phosphorylated CtrA) directly binds the origin of replication to block initiation, represses division genes, and activates flagellar genes (Quon et al., 1996; Ryan et al., 2002). CtrA~P is present in swarmer cells (preventing replication and division) but degraded in stalked cells (permitting replication) (Domian et al., 1997; Jenal, 2000). The phosphorylation cascade $PleC \rightarrow DivK \rightarrow CckA \rightarrow CtrA$ operates independently of physical cell state—a swarmer cell with sufficient size and nutrients cannot initiate replication until the molecular cascade reprograms it (Tsokos & Laub, 2012; Kott et al., 2014). The molecular hierarchy overrides permissive physical conditions.

Type B: Bidirectional Coupling—Physical and molecular systems influence each other continuously during normal homeostasis.

Canonical example: DNA supercoiling affects replication and transcription, while replication and transcription alter DNA topology (Liu & Wang, 1987; Dorman, 2013). Neither level dominates; they are coupled in a continuous bidirectional relationship.

Type C: Physical Default—Physical processes dominate when molecular regulation is minimal or absent.

Controlled Type C example: When specific cell cycle regulators are deleted in *E. coli*, division timing variability increases dramatically. For example, deletion of key division genes (e.g., *ftsZ*

temperature-sensitive mutants) or regulatory components (e.g., *min* mutants) produces cells with highly variable division timing and placement errors (Bi & Lutkenhaus, 1991; Ghosal et al., 2021; Huang et al., 2024). These well-controlled systems isolate the effect of reduced regulatory complexity on division precision.

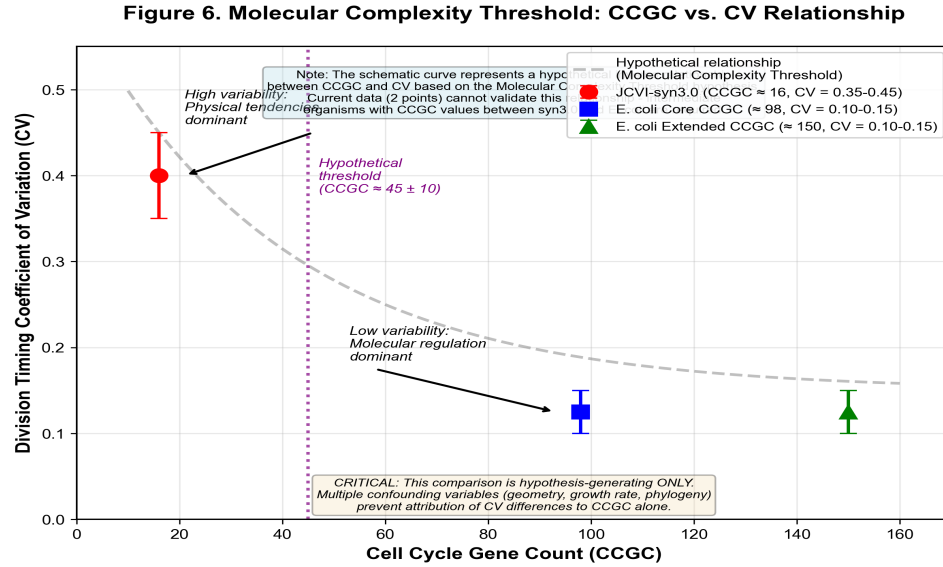


Figure 4. Type C Organization: Molecular complexity buffers against physical stochasticity. Left: In wild-type *E. coli* with full regulatory complexity, division timing is precise (low CV). Right: When key regulators are deleted, division timing becomes highly variable (high CV).

4.2 The Asymmetry Index (AsI): Operational Definition and Limitations

To operationalize the hierarchical framework, I introduce the Asymmetry Index (AsI) as a quantitative metric for assessing the relative strength of molecular versus physical influences on cell cycle outcomes:

$$\text{AsI} = |\Delta\mathbf{M}/\sigma\mathbf{M}| / |\Delta\mathbf{P}/\sigma\mathbf{P}|$$

where:

- $\Delta\mathbf{M}$ = Mean effect of molecular perturbation on cell cycle outcome
- $\Delta\mathbf{P}$ = Mean effect of physical perturbation on cell cycle outcome
- $\sigma\mathbf{M}$ = Standard deviation of molecular perturbation effect across biological replicates
- $\sigma\mathbf{P}$ = Standard deviation of physical perturbation effect across biological replicates

Interpretation: $\text{AsI} \gg 1$ indicates molecular dominance (Type A), $\text{AsI} \approx 1$ indicates bidirectional coupling (Type B), $\text{AsI} \ll 1$ indicates physical dominance (Type C).

4.2.1 Dimensionality and Comparability

A fundamental challenge is that molecular and physical perturbations are operationally incommensurable. To make $\Delta\mathbf{M}$ and $\Delta\mathbf{P}$ comparable, all perturbations must be normalized to their

physiological range:

$$\Delta M_{\text{normalized}} = \Delta M / \Delta M_{\text{max_possible}}$$

$$\Delta P_{\text{normalized}} = \Delta P / \Delta P_{\text{max_possible}}$$

where $\Delta M_{\text{max_possible}}$ is the effect size of the strongest molecular perturbation and $\Delta P_{\text{max_possible}}$ is the effect size of the strongest physical perturbation within survivable bounds.

4.2.2 Perturbation Specificity and the Circular Validation Problem

Crucially, "molecular" and "physical" perturbations are not cleanly separable—molecular perturbations have immediate physical consequences and vice versa. AsI measures EFFECTIVE pathway asymmetry in a specific context, capturing the NET effect through ALL pathways.

The circular validation problem is severe: without an independent mechanistic ground truth, ANY observed AsI value is consistent with multiple mechanisms. AsI should be viewed as a **hypothesis-generating metric** that motivates convergent multi-modal validation, NOT as a definitive discriminator.

4.2.3 Statistical Framework

The normalisation by σM and σP represents BIOLOGICAL VARIABILITY across replicates, not measurement uncertainty. High $|\Delta/\sigma|$ ratios (> 3) indicate robust, reproducible effects. Low ratios (< 1) indicate highly variable effects. AsI values should be reported with confidence intervals based on bootstrap resampling.

4.3 Convergent Multi-Modal Validation

Due to the circular validation problem, definitive mechanism discrimination requires convergent multi-modal validation:

Type I (Strongest): In vitro reconstitution with controlled components

Type II (Moderate): Multi-modal convergence combining AsI with timescale analysis, curvature sensitivity, and in vitro/in vivo comparison

Type III (Weakest): Single-modality AsI measurements alone are fundamentally ambiguous

5. Experimental Validation Roadmap

5.1 Experiment 1: Simultaneous Physical-Molecular Perturbation

Goal: Directly measure AsI by systematically varying physical parameters (crowding agents, confinement, membrane tension) and molecular components (gene knockouts, protein depletion) while measuring effects on cell cycle outcomes.

Design: Apply controlled perturbations in a model system (e.g., *E. coli* division timing). Measure effects on division timing coefficient of variation, placement errors, and fidelity. Calculate AsI from the

ratio of molecular to physical effects.

Predicted outcomes: Under checkpoint conditions (e.g., DNA damage): $AsI \gg 1$. During normal homeostasis: $AsI \approx 1$. In minimal cells: $AsI \ll 1$.

5.2 Experiment 2: Min System AsI Measurement

Goal: Test whether the Min system uses active geometric sensing or passive reaction-diffusion by measuring AsI.

Design: Reconstitute Min system in vitro with controlled physical conditions. Apply physical perturbations (membrane curvature, confinement) and molecular perturbations (protein concentrations, mutants). Measure Min pattern formation and Z-ring positioning.

5.3 Experiment 3: Molecular Complexity Threshold

Goal: Test whether adding or removing cell cycle genes systematically changes division timing variability.

Design: Start with a minimal strain and systematically add cell cycle genes. Measure effects on division timing variability, placement precision, and fitness. Identify whether there is a threshold at which division precision dramatically improves.

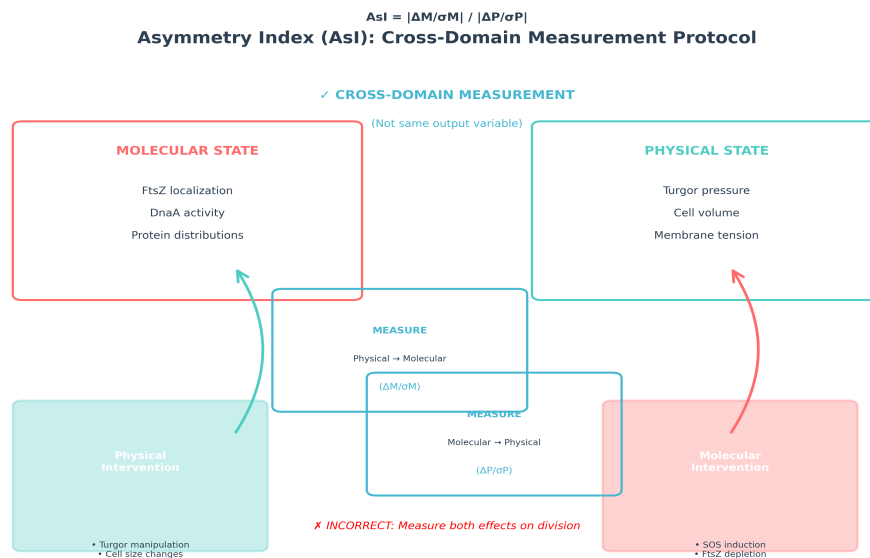


Figure 5. Asymmetry Index measurement protocol. Physical perturbations (e.g., osmotic shock, membrane tension) and molecular perturbations (e.g., gene knockouts, protein depletion) are applied while measuring cell cycle outcomes. The ratio of effects quantifies molecular versus physical influence.

6. Discussion and Conclusion

6.1 What the Framework Enables

The hierarchical framework provides: (1) conceptual integration of physical and molecular perspectives, (2) predictive power about when asymmetric information flow should evolve, (3) a quantitative metric (AsI) for assessing physical-molecular relationships, and (4) an experimental roadmap with feasible tests.

6.2 Honest Limitations

AsI circular validation problem: AsI cannot be definitively interpreted without independent knowledge of mechanisms. Definitive conclusions require convergent multi-modal validation.

Evolutionary questions: Deep-time evolutionary questions about LUCA's division mechanism remain intractable with current data.

Scope: The framework focuses on model systems (*E. coli*, *B. subtilis*). Extension to other bacteria requires further research.

6.3 Relation to Previous Work

The hierarchical framework engages with several prior quantitative frameworks. Halatek & Frey (2012) provide a detailed quantitative Min system model based on reaction-diffusion equations. **What this framework adds:** We place Min in context within a broader typology as Type B and make novel predictions about AsI values distinguishing between passive reaction-diffusion versus active geometric sensing mechanisms.

Turing (1952) provides theoretical foundations for pattern formation through reaction-diffusion systems. **What this framework adds:** We position reaction-diffusion patterns within Type B bidirectional coupling and predict when such systems should evolve preferentially.

Transfer entropy provides model-free, information-theoretic methods for measuring directed information flow. **What this framework adds:** We provide functional-contextual interpretation of information flow, predicting that transfer entropy should vary systematically across functional contexts.

Structural equation modeling (SEM) provides mathematical frameworks for inferring causal relationships from observational data. **What this framework adds:** We provide evolutionary and functional predictions about when each organizational type should be favored by natural selection.

6.3.5 Genuine Novel Contributions

Novel Contribution 1: Context-Dependent Organizational Typology

Prior frameworks focus on specific mechanisms. This framework provides a CROSS-CONTEXTUAL typology that explains WHY the same molecular machinery (e.g., FtsZ) participates in different organizational relationships in different functional contexts. FtsZ participates in Type A override during SOS checkpoint, Type B bidirectional coupling during normal homeostasis, and Type C physical-default regime in minimal cells. This CONTEXT-DEPENDENT typology is NOT provided by prior frameworks.

Novel Contribution 2: Evolutionary Predictions

The framework generates testable evolutionary predictions about which organisms should exhibit which organizational types based on retained division machinery and ecological niche. These predictions about PHYLOGENETIC VARIATION are NOT predictions made by prior frameworks.

Novel Contribution 3: Functional Logic of Organizational Type Evolution

The framework provides selective logic for WHEN each organizational type is favored by natural selection: Type A when checkpoint failure imposes severe fitness costs; Type B when continuous homeostatic management provides advantage; Type C when molecular complexity is minimal or secondarily lost.

6.4 Future Directions

Key areas for future research include: (1) AsI measurements in diverse systems, (2) comparative studies across organisms with different division mechanisms, (3) minimal cell engineering with different regulatory complexity levels, (4) methodological development for cleaner physical and molecular perturbations, and (5) investigation of Type C reversion—secondary loss of molecular complexity in obligate intracellular parasites may represent reversion to Type C physical-default organization, and whether such reversion is reversible or terminal remains an important evolutionary question.

6.5 Conclusion: Resolving the Original Question

This review began with a fundamental question: **To what extent do physical and chemical constraints contribute to bacterial cell cycle regulation, and how do these physical foundations interact with sophisticated molecular regulatory systems?**

The answer, supported by extensive existing evidence, is that bacterial cell cycle regulation is **not a matter of physical versus molecular explanations** but rather a **hierarchical integration** where the relationship between physical constraints and molecular regulation **depends on functional context**:

1. **During critical functional transitions** (checkpoints, stress responses, developmental programming), molecular regulation **hierarchically overrides** physical constraints. The SOS DNA damage checkpoint provides definitive evidence: SulA inhibits FtsZ regardless of permissive physical conditions. *Caulobacter* asymmetric division demonstrates molecular programming that overrides physical symmetry.
2. **During normal homeostasis**, physical and molecular systems exhibit **bidirectional coupling**. DNA supercoiling affects replication and transcription, while replication and transcription alter DNA topology. The Min system self-organizes through reaction-diffusion dynamics while responding to cell geometry.
3. **When molecular regulation is minimal**, physical processes create **default behaviors**. Minimal cells and in vitro reconstitution systems demonstrate that division can occur through physical processes alone, but with reduced precision and robustness.

This framework resolves the original question by providing predictive understanding of when each organizational type is favored by natural selection. Type A is favored where checkpoint failure imposes severe fitness costs. Type B is favored where continuous homeostatic management provides advantage. Type C represents the ancestral physical baseline or secondary loss of molecular complexity.

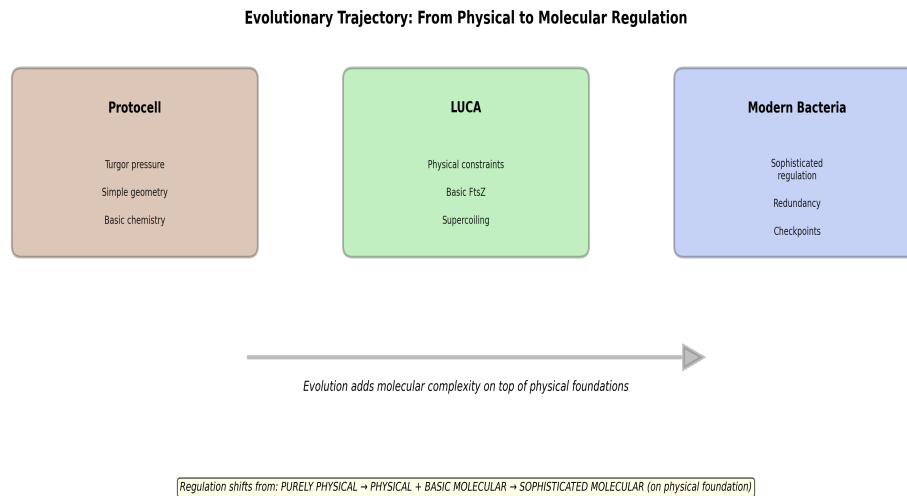


Figure 6. Evolutionary implications of the hierarchical framework. Early cells likely operated with Type C physical-default organization. As complexity increased, Type B bidirectional coupling became advantageous for homeostatic management. Type A hierarchical override was favored in lineages where precise checkpoint control provided strong selective advantage.

The framework is supported by extensive existing evidence: Type A (SOS checkpoint, *Caulobacter*), Type B (DNA supercoiling, Min system), and Type C (minimal cells, in vitro reconstitution).

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